



PNEUMATICS PLAN

MACHINE NUMBER: AS000019

MACHINE DESCRIPTION: FB FlexiBowl® 200CC/E V3.0

A	 ARS Automation - Arezzo - IT	PNEUMATIC	Disegnatore/Designer:	Antonio Esposito	Commessa/Project nr.:		A3	=	#	A	
			Approvazione/Approval:					+			
			Data/Date:	29/01/2026	Matricola/Serial nr.:			Rev.:	Page		1
			Modello macchina/Machine model:	FB_FlexiBowl® 200CC/E V3.0	Codice/Equipment code:			AS000019	Total		5
	8	7	6	5	4	3	2	1			

	8	7	6	5	4	3	2	1							
F	PLATE DATA				HOSE		PIPES			F					
E	Equipment code		AS000019		Standard hose material:		Polyamide (PA)			E					
	Machine model:		FB_FlexiBowl® 200CC/E V3.0		Special hose material:		Special pipes material:								
	Air capacity:		l/min												
Air pressure at least:		Bar													
D							Exhaust air not collected			D					
					Exhaust air collected										
RED															
BLUE					Cylinder expulsion										
BLACK					Air supply										
C					WHITE					C					
					GRAY										
					GREEN										
					YELLOW										
					CLEAR										
B							LABELING			B					
							Hose numbering:								
							ARS STANDARD								
							Components marking:								
							ARS STANDARD =MACHINE+CABINET#CATEGORY-COMPONENT								
A							Hose connection numbering:			A					
							ARS STANDARD =MACHINE+CABINET#CATEGORY-COMPONENT:CONNECTION								
ars		DATA SHEET		Disegnatore/Designer:		Antonio Esposito		Commissa/Project nr.:		A3		=		#	
				Approvazione/Approval:								+			
				Data/Date:		29/01/2026		Matricola/Serial nr.:		Codice/Equipment code:		Rev.:		Page 2	
				Modello macchina/Machine model:		FB_FlexiBowl® 200CC/E V3.0				AS000019				Total 5	
	8	7	6	5	4	3	2	1							

		8		7		6		5		4		3		2		1																																																																																																																																			
F		<table><tr><th>CODE</th><th>DESCRIPTION</th></tr><tr><td>A</td><td>Two or more purposes or tasks</td></tr><tr><td>AF</td><td>Control panel</td></tr><tr><td>B</td><td>Conversion of input variables (physical property, condition or event) in a signal intended for further processing</td></tr><tr><td>BF</td><td>Flow meter</td></tr><tr><td>BG</td><td>Position sensor, proximity switch</td></tr><tr><td>BK</td><td>Operating hours counter</td></tr><tr><td>BL</td><td>Level sensor</td></tr><tr><td>BP</td><td>Pressure sensor, vacuum circuit-breaker</td></tr><tr><td>BT</td><td>Temperature sensor, sensor</td></tr><tr><td>BX</td><td>Camera</td></tr><tr><td>C</td><td>Storing of energy, information or material</td></tr><tr><td>CA</td><td>Capacitor, capacitive buffer module</td></tr><tr><td>E</td><td>Supply of radiation or thermal energy</td></tr><tr><td>EA</td><td>Light, illumination, UV light, camera lighting</td></tr><tr><td>EB</td><td>Heating cartridge, heating rod, E-heating</td></tr><tr><td>EC</td><td>Cooling device (only cooling)</td></tr><tr><td>EZ</td><td>Temperature controllers (cooling, heating)</td></tr><tr><td>F</td><td>Direct (independent) protection of an energy or signal flow, from persons or equipment against dangerous or undesirable conditions including systems and equipment for safety purposes</td></tr><tr><td>FB</td><td>Protection against residual current (RCD - residual current device)</td></tr><tr><td>FC</td><td>Protection against excess currents (NH isolators, circuit breakers, motor protectors)</td></tr><tr><td>FN</td><td>Magnetic guard locking for safety doors (process protection)</td></tr><tr><td>FQ</td><td>Protection of individuals (safety door switch, safety light barrier)</td></tr><tr><td>FZ</td><td>Safety PLC (possibly also I/O cards??? - otherwise KF)</td></tr><tr><td>G</td><td>Initiation of energy or material flow; generation of signals used as information carriers or references sources</td></tr><tr><td>GQ</td><td>Fan, fan, vacuum pump, compressor, suction cup (dust extractor)</td></tr><tr><td>K</td><td>Processing (input, processing and output) of signals or information (except objects for safety purposes, see code F)</td></tr><tr><td>KF</td><td>Relays, optocouplers, input/output assemblies (PLC cards), regulators, PLC</td></tr><tr><td>KZ</td><td>Valve terminal (with bus connection) (incl. IO assemblies e.g. CPX)</td></tr><tr><td>M</td><td>Provision of mechanical energy (mechanical rotary or linear motion) for drive purposes</td></tr><tr><td>MA</td><td>Motor, linear motor, adjusting units, servo motors</td></tr><tr><td>MB</td><td>Electromagnet</td></tr><tr><td>P</td><td>Representation of information</td></tr><tr><td>PF</td><td>Pilot lamp</td></tr><tr><td>PJ</td><td>Horn, buzzer</td></tr><tr><td>PZ</td><td>Signal light with acoustic element</td></tr><tr><td>Q</td><td>Controlled switching or varying of energy, signal or material flow</td></tr><tr><td>QA</td><td>Main switch, contactor</td></tr><tr><td>QM</td><td>Solenoid valve, multipole valve block</td></tr><tr><td>QN</td><td>Proportional valve</td></tr><tr><td>R</td><td>Limiting or stabilising of movement or flow of energy, information or material</td></tr><tr><td>RA</td><td>Resistor, choke, diode</td></tr><tr><td>RF</td><td>Line filter (servos, FC)</td></tr><tr><td>S</td><td>Conversion of manual triggering into a signal intended for further processing</td></tr><tr><td>SF</td><td>Switch, button, selector switch, illuminated push button, and emergency stop button</td></tr><tr><td>T</td><td>Conversion of energy while maintaining the energy type, converting an existing signal while maintaining the information content, changing the form or design of a material</td></tr><tr><td>TA</td><td>Transformer, DC/DC inverter, FU</td></tr><tr><td>TB</td><td>24V power supply, servo line modules, servo motor module, semiconductor generator</td></tr><tr><td>TF</td><td>Measuring amplifier (e.g. at load cell)</td></tr><tr><td>U</td><td>Keeping objects in a defined position</td></tr><tr><td>UC</td><td>Control cabinet, terminal box, mounting plate, button box</td></tr><tr><td>V</td><td>Processing of materials or products (including pretreatment and secondary treatment)</td></tr><tr><td>VN</td><td>Printer (laser, thermal transfer, inkjet, etc.)</td></tr><tr><td>VQ</td><td>Dust extractor</td></tr><tr><td>W</td><td>Conducting or guiding energy, signals, material or products from one place to another</td></tr><tr><td>WC</td><td>Busbar energy distribution <1kV (400VAC/24VDC)</td></tr><tr><td>WD</td><td>Energy cable</td></tr><tr><td>WE</td><td>Busbar reference potential (PE / N / SH / 24M)</td></tr><tr><td>WF</td><td>BUS cables (PN,PB,Cube-Bus, Drive-Cligu)</td></tr><tr><td>WG</td><td>Control cables, measurement cables, data cables (e.g. interface cables for control signals, printer connection, scanner cable, etc.)</td></tr><tr><td>X</td><td>Connecting objects</td></tr><tr><td>XD</td><td>Terminal board, plug, socket (<1kV) e.g. power plug, clamping strips for energy distribution, combined clamping and plug strips</td></tr><tr><td>XE</td><td>PE disconnect terminal</td></tr><tr><td>XG</td><td>Terminals, plug-in connections (data, signals) e.g. bus plugs, interface terminal board</td></tr><tr><td>XF</td><td>Hub, switch</td></tr></table>														CODE	DESCRIPTION	A	Two or more purposes or tasks	AF	Control panel	B	Conversion of input variables (physical property, condition or event) in a signal intended for further processing	BF	Flow meter	BG	Position sensor, proximity switch	BK	Operating hours counter	BL	Level sensor	BP	Pressure sensor, vacuum circuit-breaker	BT	Temperature sensor, sensor	BX	Camera	C	Storing of energy, information or material	CA	Capacitor, capacitive buffer module	E	Supply of radiation or thermal energy	EA	Light, illumination, UV light, camera lighting	EB	Heating cartridge, heating rod, E-heating	EC	Cooling device (only cooling)	EZ	Temperature controllers (cooling, heating)	F	Direct (independent) protection of an energy or signal flow, from persons or equipment against dangerous or undesirable conditions including systems and equipment for safety purposes	FB	Protection against residual current (RCD - residual current device)	FC	Protection against excess currents (NH isolators, circuit breakers, motor protectors)	FN	Magnetic guard locking for safety doors (process protection)	FQ	Protection of individuals (safety door switch, safety light barrier)	FZ	Safety PLC (possibly also I/O cards??? - otherwise KF)	G	Initiation of energy or material flow; generation of signals used as information carriers or references sources	GQ	Fan, fan, vacuum pump, compressor, suction cup (dust extractor)	K	Processing (input, processing and output) of signals or information (except objects for safety purposes, see code F)	KF	Relays, optocouplers, input/output assemblies (PLC cards), regulators, PLC	KZ	Valve terminal (with bus connection) (incl. IO assemblies e.g. CPX)	M	Provision of mechanical energy (mechanical rotary or linear motion) for drive purposes	MA	Motor, linear motor, adjusting units, servo motors	MB	Electromagnet	P	Representation of information	PF	Pilot lamp	PJ	Horn, buzzer	PZ	Signal light with acoustic element	Q	Controlled switching or varying of energy, signal or material flow	QA	Main switch, contactor	QM	Solenoid valve, multipole valve block	QN	Proportional valve	R	Limiting or stabilising of movement or flow of energy, information or material	RA	Resistor, choke, diode	RF	Line filter (servos, FC)	S	Conversion of manual triggering into a signal intended for further processing	SF	Switch, button, selector switch, illuminated push button, and emergency stop button	T	Conversion of energy while maintaining the energy type, converting an existing signal while maintaining the information content, changing the form or design of a material	TA	Transformer, DC/DC inverter, FU	TB	24V power supply, servo line modules, servo motor module, semiconductor generator	TF	Measuring amplifier (e.g. at load cell)	U	Keeping objects in a defined position	UC	Control cabinet, terminal box, mounting plate, button box	V	Processing of materials or products (including pretreatment and secondary treatment)	VN	Printer (laser, thermal transfer, inkjet, etc.)	VQ	Dust extractor	W	Conducting or guiding energy, signals, material or products from one place to another	WC	Busbar energy distribution <1kV (400VAC/24VDC)	WD	Energy cable	WE	Busbar reference potential (PE / N / SH / 24M)	WF	BUS cables (PN,PB,Cube-Bus, Drive-Cligu)	WG	Control cables, measurement cables, data cables (e.g. interface cables for control signals, printer connection, scanner cable, etc.)	X	Connecting objects	XD	Terminal board, plug, socket (<1kV) e.g. power plug, clamping strips for energy distribution, combined clamping and plug strips	XE	PE disconnect terminal	XG	Terminals, plug-in connections (data, signals) e.g. bus plugs, interface terminal board	XF	Hub, switch	F	
CODE	DESCRIPTION																																																																																																																																																		
A	Two or more purposes or tasks																																																																																																																																																		
AF	Control panel																																																																																																																																																		
B	Conversion of input variables (physical property, condition or event) in a signal intended for further processing																																																																																																																																																		
BF	Flow meter																																																																																																																																																		
BG	Position sensor, proximity switch																																																																																																																																																		
BK	Operating hours counter																																																																																																																																																		
BL	Level sensor																																																																																																																																																		
BP	Pressure sensor, vacuum circuit-breaker																																																																																																																																																		
BT	Temperature sensor, sensor																																																																																																																																																		
BX	Camera																																																																																																																																																		
C	Storing of energy, information or material																																																																																																																																																		
CA	Capacitor, capacitive buffer module																																																																																																																																																		
E	Supply of radiation or thermal energy																																																																																																																																																		
EA	Light, illumination, UV light, camera lighting																																																																																																																																																		
EB	Heating cartridge, heating rod, E-heating																																																																																																																																																		
EC	Cooling device (only cooling)																																																																																																																																																		
EZ	Temperature controllers (cooling, heating)																																																																																																																																																		
F	Direct (independent) protection of an energy or signal flow, from persons or equipment against dangerous or undesirable conditions including systems and equipment for safety purposes																																																																																																																																																		
FB	Protection against residual current (RCD - residual current device)																																																																																																																																																		
FC	Protection against excess currents (NH isolators, circuit breakers, motor protectors)																																																																																																																																																		
FN	Magnetic guard locking for safety doors (process protection)																																																																																																																																																		
FQ	Protection of individuals (safety door switch, safety light barrier)																																																																																																																																																		
FZ	Safety PLC (possibly also I/O cards??? - otherwise KF)																																																																																																																																																		
G	Initiation of energy or material flow; generation of signals used as information carriers or references sources																																																																																																																																																		
GQ	Fan, fan, vacuum pump, compressor, suction cup (dust extractor)																																																																																																																																																		
K	Processing (input, processing and output) of signals or information (except objects for safety purposes, see code F)																																																																																																																																																		
KF	Relays, optocouplers, input/output assemblies (PLC cards), regulators, PLC																																																																																																																																																		
KZ	Valve terminal (with bus connection) (incl. IO assemblies e.g. CPX)																																																																																																																																																		
M	Provision of mechanical energy (mechanical rotary or linear motion) for drive purposes																																																																																																																																																		
MA	Motor, linear motor, adjusting units, servo motors																																																																																																																																																		
MB	Electromagnet																																																																																																																																																		
P	Representation of information																																																																																																																																																		
PF	Pilot lamp																																																																																																																																																		
PJ	Horn, buzzer																																																																																																																																																		
PZ	Signal light with acoustic element																																																																																																																																																		
Q	Controlled switching or varying of energy, signal or material flow																																																																																																																																																		
QA	Main switch, contactor																																																																																																																																																		
QM	Solenoid valve, multipole valve block																																																																																																																																																		
QN	Proportional valve																																																																																																																																																		
R	Limiting or stabilising of movement or flow of energy, information or material																																																																																																																																																		
RA	Resistor, choke, diode																																																																																																																																																		
RF	Line filter (servos, FC)																																																																																																																																																		
S	Conversion of manual triggering into a signal intended for further processing																																																																																																																																																		
SF	Switch, button, selector switch, illuminated push button, and emergency stop button																																																																																																																																																		
T	Conversion of energy while maintaining the energy type, converting an existing signal while maintaining the information content, changing the form or design of a material																																																																																																																																																		
TA	Transformer, DC/DC inverter, FU																																																																																																																																																		
TB	24V power supply, servo line modules, servo motor module, semiconductor generator																																																																																																																																																		
TF	Measuring amplifier (e.g. at load cell)																																																																																																																																																		
U	Keeping objects in a defined position																																																																																																																																																		
UC	Control cabinet, terminal box, mounting plate, button box																																																																																																																																																		
V	Processing of materials or products (including pretreatment and secondary treatment)																																																																																																																																																		
VN	Printer (laser, thermal transfer, inkjet, etc.)																																																																																																																																																		
VQ	Dust extractor																																																																																																																																																		
W	Conducting or guiding energy, signals, material or products from one place to another																																																																																																																																																		
WC	Busbar energy distribution <1kV (400VAC/24VDC)																																																																																																																																																		
WD	Energy cable																																																																																																																																																		
WE	Busbar reference potential (PE / N / SH / 24M)																																																																																																																																																		
WF	BUS cables (PN,PB,Cube-Bus, Drive-Cligu)																																																																																																																																																		
WG	Control cables, measurement cables, data cables (e.g. interface cables for control signals, printer connection, scanner cable, etc.)																																																																																																																																																		
X	Connecting objects																																																																																																																																																		
XD	Terminal board, plug, socket (<1kV) e.g. power plug, clamping strips for energy distribution, combined clamping and plug strips																																																																																																																																																		
XE	PE disconnect terminal																																																																																																																																																		
XG	Terminals, plug-in connections (data, signals) e.g. bus plugs, interface terminal board																																																																																																																																																		
XF	Hub, switch																																																																																																																																																		
E		<table><tr><td>B</td><td>Conversion of input variables (physical property, condition or event) in a signal intended for further processing</td></tr><tr><td>BP</td><td>Pressure sensor, vacuum circuit-breaker</td></tr><tr><td>BK</td><td>Time switch timing module</td></tr><tr><td>C</td><td>Storing of energy, information or material</td></tr><tr><td>CM</td><td>Storage, vessel</td></tr><tr><td>FL</td><td>Relief valve</td></tr><tr><td>DP</td><td>Radiator, injector , ejector nozzle</td></tr><tr><td>GQ</td><td>Compressor, vacuum pump, suction cup</td></tr><tr><td>HQ</td><td>Filter</td></tr><tr><td>K</td><td>Processing (input, processing and output) of signals or information (except objects for safety purposes, see code F)</td></tr><tr><td>KH</td><td>Pneumatic valve, pre-control valve, valve block</td></tr><tr><td>KZ</td><td>Valve terminal (with bus connection)</td></tr><tr><td>M</td><td>Provision of mechanical energy (mechanical rotary or linear motion) for drive purposes</td></tr><tr><td>MM</td><td>Cylinder</td></tr><tr><td>P</td><td>Representation of information</td></tr><tr><td>PH</td><td>Visual representation of individual variables, manometer, displays, vacuum meter...</td></tr><tr><td>Q</td><td>Controlled switching or varying of energy, signal or material flow</td></tr><tr><td>QM</td><td>Solenoid valve, multipole valve block</td></tr><tr><td>QN</td><td>Pressure regulator, proportional valve</td></tr><tr><td>QZ</td><td>Service unit, logic components</td></tr><tr><td>R</td><td>Limiting or stabilising of movement or flow of energy, information or material</td></tr><tr><td>RA</td><td>Choke</td></tr><tr><td>RL</td><td>Stop valve</td></tr><tr><td>RN</td><td>Venturi nozzle</td></tr><tr><td>RM</td><td>Check valve / rapid venting</td></tr><tr><td>RP</td><td>Silencer</td></tr><tr><td>RZ</td><td>One-way restrictor / one-way restrictor with stop valve</td></tr><tr><td>SJ</td><td>Manual valve</td></tr><tr><td>WN</td><td>Conduct and guide flow of liquid, flowable and gaseous materials (closed, flexible containment</td></tr><tr><td>WP</td><td>Conduct and guide flow of liquid, flowable and gaseous materials (closed, rigid containment</td></tr><tr><td>X</td><td>Connecting objects</td></tr><tr><td>XL</td><td>Flanges, fittings, couplings</td></tr><tr><td>XM</td><td>Hose connection, hose coupling</td></tr></table>														B	Conversion of input variables (physical property, condition or event) in a signal intended for further processing	BP	Pressure sensor, vacuum circuit-breaker	BK	Time switch timing module	C	Storing of energy, information or material	CM	Storage, vessel	FL	Relief valve	DP	Radiator, injector , ejector nozzle	GQ	Compressor, vacuum pump, suction cup	HQ	Filter	K	Processing (input, processing and output) of signals or information (except objects for safety purposes, see code F)	KH	Pneumatic valve, pre-control valve, valve block	KZ	Valve terminal (with bus connection)	M	Provision of mechanical energy (mechanical rotary or linear motion) for drive purposes	MM	Cylinder	P	Representation of information	PH	Visual representation of individual variables, manometer, displays, vacuum meter...	Q	Controlled switching or varying of energy, signal or material flow	QM	Solenoid valve, multipole valve block	QN	Pressure regulator, proportional valve	QZ	Service unit, logic components	R	Limiting or stabilising of movement or flow of energy, information or material	RA	Choke	RL	Stop valve	RN	Venturi nozzle	RM	Check valve / rapid venting	RP	Silencer	RZ	One-way restrictor / one-way restrictor with stop valve	SJ	Manual valve	WN	Conduct and guide flow of liquid, flowable and gaseous materials (closed, flexible containment	WP	Conduct and guide flow of liquid, flowable and gaseous materials (closed, rigid containment	X	Connecting objects	XL	Flanges, fittings, couplings	XM	Hose connection, hose coupling	E																																																																	
B	Conversion of input variables (physical property, condition or event) in a signal intended for further processing																																																																																																																																																		
BP	Pressure sensor, vacuum circuit-breaker																																																																																																																																																		
BK	Time switch timing module																																																																																																																																																		
C	Storing of energy, information or material																																																																																																																																																		
CM	Storage, vessel																																																																																																																																																		
FL	Relief valve																																																																																																																																																		
DP	Radiator, injector , ejector nozzle																																																																																																																																																		
GQ	Compressor, vacuum pump, suction cup																																																																																																																																																		
HQ	Filter																																																																																																																																																		
K	Processing (input, processing and output) of signals or information (except objects for safety purposes, see code F)																																																																																																																																																		
KH	Pneumatic valve, pre-control valve, valve block																																																																																																																																																		
KZ	Valve terminal (with bus connection)																																																																																																																																																		
M	Provision of mechanical energy (mechanical rotary or linear motion) for drive purposes																																																																																																																																																		
MM	Cylinder																																																																																																																																																		
P	Representation of information																																																																																																																																																		
PH	Visual representation of individual variables, manometer, displays, vacuum meter...																																																																																																																																																		
Q	Controlled switching or varying of energy, signal or material flow																																																																																																																																																		
QM	Solenoid valve, multipole valve block																																																																																																																																																		
QN	Pressure regulator, proportional valve																																																																																																																																																		
QZ	Service unit, logic components																																																																																																																																																		
R	Limiting or stabilising of movement or flow of energy, information or material																																																																																																																																																		
RA	Choke																																																																																																																																																		
RL	Stop valve																																																																																																																																																		
RN	Venturi nozzle																																																																																																																																																		
RM	Check valve / rapid venting																																																																																																																																																		
RP	Silencer																																																																																																																																																		
RZ	One-way restrictor / one-way restrictor with stop valve																																																																																																																																																		
SJ	Manual valve																																																																																																																																																		
WN	Conduct and guide flow of liquid, flowable and gaseous materials (closed, flexible containment																																																																																																																																																		
WP	Conduct and guide flow of liquid, flowable and gaseous materials (closed, rigid containment																																																																																																																																																		
X	Connecting objects																																																																																																																																																		
XL	Flanges, fittings, couplings																																																																																																																																																		
XM	Hose connection, hose coupling																																																																																																																																																		
D																D																																																																																																																																			
C																C																																																																																																																																			
B																B																																																																																																																																			
A																A																																																																																																																																			

[illegible]

